

KATOWICE MUCHOWEC, Poland

WMO: 125600

Lat: 50.2406N

Lon: 19.0328E

Elev: 920

StdP: 14.21

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	10.9	0.5	5.8	6.2	6.0	7.7	12.6	23.1	40.9	20.5	39.0	3.5	60	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	86.2	68.0	82.6	66.3	79.5	65.0	69.7	81.5	68.2	79.3	66.7	76.8	7.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	99.1	74.1	64.4	93.8	72.2	62.9	88.8	70.8	34.2	81.5	32.9	79.9	31.7	77.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.2	14.1	-1.2	91.4	7.1	2.7	-6.3	93.3	-10.4	94.9	-14.4	96.4	-19.6	98.4		
			-1.7	73.1	7.0	1.6	-6.7	74.2	-10.8	75.2	-14.7	76.1	-19.8	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	29.1	31.9	38.1	48.1	56.3	62.4	65.9	65.0	56.6	48.1	39.8	31.6
	DBStd	14.98	9.45	9.25	7.88	7.69	6.79	6.32	5.59	5.93	6.21	7.48	7.76	8.74
	HDD50	2698	648	508	373	124	19	1	0	0	12	128	315	570
	HDD65	6491	1113	928	833	509	279	122	57	74	260	524	757	1035
	CDD50	1906	0	0	5	65	214	374	494	466	210	70	8	0
	CDD65	225	0	0	0	1	9	46	87	75	7	0	0	0
	CDH74	2387	0	0	0	25	156	490	906	720	88	2	0	0
	CDH80	613	0	0	0	2	18	118	263	199	13	0	0	0
	WSAvg	6.0	7.1	7.0	7.0	6.1	5.4	5.4	5.2	4.8	5.2	5.7	6.3	7.1
	PrecAvg	30.6	1.8	1.7	2.0	2.0	3.3	3.6	4.2	3.1	2.9	2.1	2.1	1.7
Precipitation	PrecMax	43.5	4.2	4.1	4.4	4.3	12.4	7.4	11.8	6.5	6.5	4.3	3.9	5.2
	PrecMin	23.0	0.6	0.5	0.8	0.0	0.9	1.4	0.4	0.5	0.7	0.2	0.0	0.6
	PrecStd	4.7	0.8	0.9	0.9	0.9	2.1	1.5	2.3	1.4	1.4	1.2	0.9	0.9
	PrecStd	4.7	0.8	0.9	0.9	0.9	2.1	1.5	2.3	1.4	1.4	1.2	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	55.5	64.7	77.3	82.3	88.2	90.6	90.8	81.3	72.5	63.1	53.9
		MCWB	45.8	46.7	50.2	58.4	63.0	68.5	69.0	68.5	64.0	60.3	54.0	47.6
	2%	DB	47.1	51.1	59.2	71.8	78.5	83.7	86.4	85.2	76.2	68.4	58.8	48.5
		MCWB	43.4	44.4	48.0	55.6	62.0	67.1	67.8	67.9	63.0	58.4	51.9	44.5
	5%	DB	44.1	47.8	55.1	67.1	74.6	79.5	82.6	81.3	72.5	64.3	55.0	45.6
		MCWB	41.1	42.9	46.4	53.1	60.3	64.8	66.7	66.8	61.2	56.9	50.1	42.6
	10%	DB	41.3	44.3	50.9	62.9	70.8	76.1	79.1	77.8	68.9	60.1	51.6	42.8
		MCWB	38.7	40.4	44.2	51.1	58.5	63.3	65.3	65.0	59.7	54.9	47.8	40.5
	0.4%	WB	46.8	47.9	51.8	59.4	66.1	70.8	72.0	71.4	66.5	62.0	54.9	48.5
		MCDB	50.1	52.8	61.1	74.0	78.4	83.0	84.5	82.9	77.2	69.6	60.7	52.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.6	45.4	49.0	56.7	63.5	68.5	69.9	69.3	64.3	59.4	52.6	45.0
		MCDB	46.4	50.1	57.3	68.8	74.5	79.8	81.6	81.1	72.5	66.0	57.7	48.0
	5%	WB	41.2	43.1	47.0	54.5	61.5	66.6	68.2	67.8	62.6	57.4	50.3	42.7
		MCDB	43.6	47.4	53.7	64.7	71.8	77.1	79.2	78.8	70.3	63.1	54.4	45.0
	10%	WB	38.9	40.5	44.8	52.3	59.7	64.6	66.7	66.2	60.7	55.2	48.0	40.6
		MCDB	41.2	43.9	50.2	61.2	68.8	73.8	76.5	75.7	67.6	59.9	51.1	42.7
Mean Daily Temperature Range	5% DB	MDBR	9.9	11.9	15.1	19.5	19.9	19.1	19.5	19.9	17.7	15.8	11.8	9.6
		MCDBR	11.9	16.9	23.6	27.5	26.8	25.3	26.3	25.6	24.6	22.7	18.9	12.0
		MCWBR	9.2	11.5	14.3	14.0	12.5	11.5	11.1	11.1	12.8	13.7	12.7	9.1
	5% WB	MCDBR	11.2	14.7	20.9	23.8	23.6	22.5	23.4	22.8	21.9	20.8	16.8	11.2
		MCWBR	9.0	10.6	13.3	12.9	11.8	10.8	10.6	10.4	12.0	13.0	11.9	9.0
Clear-Sky Solar Irradiance	taub		0.315	0.342	0.382	0.434	0.428	0.428	0.445	0.436	0.405	0.372	0.346	0.313
	taud		2.287	2.267	2.221	2.141	2.203	2.237	2.200	2.230	2.288	2.351	2.364	2.351
	Ebn at Noon		223	248	257	256	263	264	257	252	248	234	211	208
	Edh at Noon		22	29	36	44	43	42	43	40	34	27	21	18
All-Sky Solar Radiation	RadAvg		265	498	852	1285	1561	1654	1590	1429	1018	609	300	206
	RadStd		47	83	130	161	192	170	197	123	147	87	42	26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.81	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	+56
(47)	Regional (37 neighbors)	+0.92	N/A	N/A	+1.42	+0.70	+0.63	N/A	-293	+121	+47

Nomenclature: See separate page